Supplementary materials and methods

Mathematical definition of StaBiCut stability metrics and scoring components

StaBiCut (Stable and Biologically aware Cutoff prioritization) was implemented as a deterministic, direction-aware framework for survival cutoff selection and post-selection prioritization. In the current implementation, the pipeline operates on log2(TPM+1)-transformed expression data, removes genes with very low mean expression, restricts survival analyses to tumor samples, and performs per-gene cutoff analysis on winsorized tumor expression values to limit the influence of extreme outliers. For each gene, a per-sample analysis table $D = \{(xi, ti, \delta_i)\}_{i=1}^{n}$ was constructed, where xi denotes winsorized tumor expression, ti denotes follow-up time in months, and δ_i denotes the event indicator. Tumor-normal comparisons were computed from the full log2(TPM+1) expression values in tumor and normal samples rather than the winsorized tumor-only analysis table.

An optional direction-prior layer was applied before cutpoint scanning. Let

$$\Delta_{TN} = median(x^{tumor}) - median(x^{normal}),$$

denote the tumor−normal median difference, where x^{tumor} and x^{normal} denote the full log2(TPM+1) expression values in tumor and normal samples, respectively. Let β_{cont} denote the coefficient from the continuous Cox model

$$h(t \mid x) = h_0(t)\exp(\beta_{cont}x).$$

In the current code path, expected direction was determined in the following order. First, if a gene-specific prior table contained a valid expected direction, that value was used directly. In the present study, all 11 candidate genes had been pre-nominated from longitudinal transcriptomic and proteomic analyses, so gene-specific directional priors were available before survival stratification. The cutoff scan was accordingly constrained by the expected directions summarized in Direction prior summary of online supplementary table 7, with adverse_high assigned to candidates showing progressive upregulation and protective_high assigned to candidates showing progressive downregulation in the CAC multi-omics profiles. More generally, when such prior information is unavailable, StaBiCut can infer expected direction only when the tumor-normal shift and the sign of the continuous Cox coefficient are concordant; otherwise, no direction constraint is imposed. Second, only if no valid gene-specific prior was available, fallback inference was allowed when Δ_{TN} and β_{cont} were both available and concordant: $\Delta_{TN} < 0$ and $\beta_{cont} < 0$ implied protective_high, whereas $\Delta_{TN} > 0$ and $\beta_{cont} > 0$ implied adverse_high. Third, a force direction branch was implemented as a stress-testing mode based on the sign of Δ_{TN} alone, but this branch was not used in the present analysis. This fallback rule was implemented in the StaBiCut workflow but was not used in the present 11-gene analysis, which relied on explicit gene-specific priors.

Candidate cutoffs were then scanned deterministically within the central tumor-expression range defined by the minimum group proportion parameter minprop m (default $m = 0.25$). Let $Q_p(x)$ denote the empirical p-th quantile of the winsorized tumor-expression vector x. The admissible search interval was therefore

$$(Q_m(x), Q_{1-m}(x)),$$

with minprop = 0.25 in the current runner.

With the current runner, candidate cutoffs were generated on a quantile grid within this interval. When grid = "quantile", candidate cutoffs were generated as unique quantile-grid values inside this admissible interval rather than by continuous optimization. Accordingly, candidate cutoffs were evaluated only between the 25th and 75th percentiles of the tumor-expression distribution. Rather than performing continuous optimization over this interval, the primary scan assessed a fixed set of quantile-based candidate values, generated at a resolution corresponding to the default grid density of 200 and then collapsed to unique admissible cutoffs within the allowed range.

For each candidate cutoff c, samples were dichotomized into

$$gi(c) = \begin{Bmatrix} \text{High}, \ xi > c, \\ \text{Low}, xi \leq c. \end{Bmatrix}$$

A candidate was discarded if it violated any feasibility criterion: minimum group proportion $< m$, fewer than 5 events in either group, failed Cox fitting, non-finite coefficients, non-finite standard errors, $se > 3$, or $\mid \beta_c \mid > 5$, where β_c is the dichotomized Cox coefficient. These latter filters were implemented as practical guards against unstable edge-peak or separation-like solutions. If expected_dir was not NULL, an additional direction constraint was imposed: $HR(c) < 1$ was required for protective_high, and $HR(c) > 1$ was required for adverse_high.

Among the remaining valid candidates, the selected cutoff c^* was chosen by a fixed lexicographic ordering rule. Under the primary cutoff-selection rule used in the present analysis (stored internally as max_wald_then_min_p), candidates were ordered by decreasing score, then increasing Wald-test P value, and finally by increasing $\mid HR(c) - 1 \mid$ as a final deterministic tie-breaking criterion. Under the current primary runner, score equals the Wald statistic because $\lambda = 0$ and no reference-cutoff penalty is applied. Thus, candidate cutoffs were prioritized by the largest Wald statistic, followed by the smallest Wald-test P value and, if still tied, the smallest absolute deviation of the hazard ratio from 1. This rule makes cutoff selection deterministic for a fixed dataset and decision rule rather than dependent on arbitrary tie resolution.

After selection of c^*, the final dichotomized survival association was estimated on the full per-gene tumor dataset D using $Surv(time, event) \sim I(expr\ >\ c*)$, from which the hazard ratio $HR^* = exp(\beta^*)$, Wald-test P value, and 95% confidence interval were extracted. A log-rank test at the same cutoff was also computed. Across genes, both the cutoff-based Cox P values and the log-rank P values were adjusted by the Benjamini–Hochberg method in the current runner.

To assess the internal stability of the selected cutoff under bootstrap resampling, StaBiCut performed bootstrap resampling with replacement and rescanned candidate cutoffs only within a local neighborhood around c^* rather than re-optimizing over the entire expression range.

When local restriction was enabled, bootstrap resampling was designed as a local reproducibility test for the originally selected cutoff within its immediate expression neighborhood, rather than as repeated de novo optimization over the full expression range. The local search window was centered on the selected cutoff and scaled to the tumor-expression interquartile range. Specifically, if IQR_x denotes the interquartile range of the winsorized tumor-expression vector in the original dataset, the bootstrap candidate window was defined as

$$[c^* - \omega IQR_x , c^* + \omega IQR_x],$$

where ω was set to 0.5, corresponding to half of the interquartile range in the current implementation.

Accordingly, candidate cutoffs within each bootstrap replicate $D^{(b)}$ were proposed only from this local search interval, using a quantile-based grid of lower density than that used in the primary scan (bootstrap grid density 120; primary grid density 200; minprop = 0.25; minimum 5 events per group). Once a bootstrap-selected cutoff $\widetilde{c_b}$ had been proposed, however, its hazard direction was re-evaluated on the full bootstrap sample before retention. Thus, local restriction constrained only the proposal of candidate cutoffs, whereas direction-consistent support still had to be demonstrated at the level of the complete resampled cohort.

Let $B = \{\widetilde{c_b}\}_{b=1}^{n_{valid}}$ denote the set of valid retained bootstrap cutoffs. Bootstrap variability was characterized by its median, interquartile range,

$$IQR\_boot = Q_{0.75}(B) - Q_{0.25}(B),$$

and empirical 95% interval width,

$$CI\_boot^{95} = Q_{0.975}(B) - Q_{0.025}(B).$$

In the current implementation, however, these bootstrap summary statistics were considered interpretable only when at least 50 valid bootstrap cutoffs were retained; otherwise, they were treated as unavailable for downstream use.

To compare genes on a common relative scale, both the bootstrap IQR and the empirical central 95% percentile interval width were normalized by the interquartile range of the original winsorized tumor-expression vector x:

$$IQR_{rel} = \frac{IQR_{boot}}{IQR_x}, \quad CI_{rel} = \frac{CI_{boot}^{95}}{IQR_x}.$$

The bootstrap stability component was then defined as

$$S_{boot} = exp(-k_1 IQR_{rel}) exp(-k_2 CI_{rel}),$$

with $k_1 = 3$ and $k_2 = 2$ in the current implementation. Narrower bootstrap dispersion relative to the intrinsic expression scale therefore yields a larger stability score. If either the bootstrap summary or IQR_x was unavailable or non-finite, this component was returned as missing.

Post-selection cutoff-position metrics were then computed. In the current implementation, this cutoff-plausibility score is stored internally as density_score for historical code continuity. First, cutoff centrality within the tumor-expression distribution was quantified by the empirical cumulative distribution function

$$q^* = F_x(c^*) = \frac{1}{n} \sum_{i=1}^{n} \boldsymbol{I}\,(x_i \leq c^*).$$

The corresponding centrality score was defined as

$$S_{plausibility} = 1 - |q^* - 0.5|.$$

This quantity is highest for cutoffs near the distributional center and lower for cutoffs closer to the tails. Although retained in the code as density_score, this ECDF-based metric measures distributional centrality rather than kernel density.

Second, group balance was quantified from the final dichotomization at c^*. If n_{low} and n_{high} denote the sizes of the Low and High groups, respectively, then

$$p_{min} = \frac{\min(n_{low}, n_{high})}{n}.$$

The balance component used in the composite score was

$$S_{balance} = \min\{1, \max(0, 2p_{min})\}$$

.

Because candidate generation had already enforced the minimum group proportion constraint, this term acts as a post-selection soft preference among admissible cutoffs rather than as the primary feasibility filter itself.

Third, tumor-normal directional concordance was evaluated by comparing the sign of the tumor-normal shift with the sign of the cutoff-based hazard direction:

$$C_{TN} = \operatorname{sign}(\Delta_{TN}) \times \operatorname{sign}\{\log(HR^*)\}.$$

Accordingly, $C_{TN} = 1$ indicates directional concordance and $C_{TN} = -1$ indicates directional discordance. If Δ_{TN} was unavailable, HR^* was unavailable or non-positive, or either sign was zero, this term was treated as missing. For composite scoring, the current runner maps this quantity to

$$S_{TN} = \begin{Bmatrix} 1, & C_{TN} = 1, \\ 0, & C_{TN} = -1, \\ NA, & \text{otherwise.} \end{Bmatrix}$$

Thus, under the current implementation, missing directional evidence propagates to this component rather than being replaced by an intermediate value.

Fourth, bootstrap hazard-direction consistency was used to assess whether the retained bootstrap-supported cutoffs preserved the same survival orientation as the reference cutoff. Let

$$d_0 = \operatorname{sign}\{\log(HR^*)\},$$

denote the reference hazard direction on the original tumor dataset D. For each retained bootstrap cutoff $\widetilde{c_b}$ in B, let

$$d_b = \operatorname{sign}\{\log(HR(\widetilde{c_b}; D))\},$$

where $HR(\widetilde{c_b}; D)$ denotes the hazard ratio recomputed on the original tumor dataset D at cutoff $\widetilde{c_b}$. Bootstrap hazard-direction consistency was then defined as the proportion of retained bootstrap cutoffs for which $d_b = d_0$, then

$$S_{hazard} = \frac{1}{M}\sum_{b=1}^{M} \boldsymbol{I}\,(d_b = d_0),$$

where M is the number of bootstrap cutoffs yielding valid direction estimates on the original dataset D. In the current implementation, this quantity was returned as missing if fewer than 30 valid bootstrap cutoffs were retained, if the reference hazard direction could not be defined, or if fewer than 30 retained bootstrap cutoffs yielded evaluable hazard directions after re-estimation on the original dataset.

The overall composite prioritization score integrated the five post-selection components using fixed weights:

$$S_{composite} = 0.40\, S_{boot} + 0.20\, S_{plausibility} + 0.15\, S_{hazard} + 0.15\, S_{TN} + 0.10\, S_{balance}.$$

By construction, the bootstrap stability component receives the largest weight (0.40) in the current implementation. Genes were ranked by descending $S_{composite}$, with minimum-rank assignment for ties. In the current runner, the composite score is computed directly from these five components, with no explicit handling of missing component values; accordingly, missing values in any required component can lead to an unavailable composite score.

Taken together, StaBiCut should be interpreted as a prior-aware, direction-constrained framework for prioritizing survival cutoffs, rather than as a purely data-driven search for the most statistically significant threshold. Its main mathematical contribution lies not in proposing a new survival test, but in formally integrating deterministic candidate filtering, local bootstrap support, direction-aware plausibility assessment, and fixed-weight post-selection ranking into a coherent prioritization strategy. Under this design, its selected cutoffs are intended to provide robust working thresholds for cohort-level stratification under the stated decision rules, not as exact biological boundaries.

Mathematical formulation of the external Meta-TN extension

The external Meta-TN extension was used to provide an independently estimated tumor-normal biological-support component when StaBiCut was applied to the tumor-only GSE39582[1] survival cohort. It extends, rather than identically re-estimates, the tumor-normal component used in the TCGA discovery analysis. The discovery component recorded directional agreement as a binary score, whereas the external Meta-TN component provides a graded score between 0 and 1 based on the reproducibility, magnitude, and meta-analytic support of tumor-normal expression differences across independent cohorts. The Meta-TN component occupied the original tumor-normal weight of 0.15 in the five-component StaBiCut score. Its addition did not change the GSE39582[1] expression preprocessing, cutoff scan, selected cutoff, bootstrap resampling, cutoff-plausibility score, bootstrap hazard-direction consistency, or group-balance score. It affected only the final external composite score and the resulting gene ranking.

Cohort-level preprocessing and sample definition

Three independent colorectal cancer datasets containing primary tumor and normal colorectal mucosa were analysed: GSE21510[2], GSE32323[3], and GSE41258[4]. Each dataset was processed separately, and expression matrices from different studies or Affymetrix platforms were not pooled. Probe identifiers were mapped to gene symbols using platform-specific annotation resources (GPL570: *hgu133plus2.db*; GPL96: *hgu133a.db*). When multiple probes mapped to the same gene, the probe with the highest median expression across all samples in that cohort was retained.

For each dataset, predefined terms in the GEO sample annotations were used to classify samples as primary colorectal tumor, normal colorectal mucosa, or excluded. Samples matching an exclusion term were removed first. Samples matching only tumor terms or only normal terms were assigned accordingly. When both sets of terms were present, an explicit “normal” annotation took priority; otherwise, the sample was classified as tumor. All sample assignments and the annotation text used for classification were exported for manual review. Tumor and normal samples were analysed as two groups. Even when a dataset contained matched specimens from the same patient, the implemented calculation did not use within-patient expression differences.

Cohort-specific standardized tumor–normal effects

For gene j in cohort k, let $\bar{x}_{T,jk}$ and $\bar{x}_{N,jk}$ denote the mean expression values in tumor and normal samples, respectively; $n_{T,jk}$ and $n_{N,jk}$ their sample sizes; and $s_{T,jk}$ and $s_{N,jk}$ their within-group standard deviations. The pooled within-cohort standard deviation was

$$s_{p,jk} = \sqrt{\frac{(n_{T,jk}-1)s_{T,jk}^2 + (n_{N,jk}-1)s_{N,jk}^2}{n_{T,jk}+n_{N,jk}-2}}.$$

The uncorrected standardized mean difference was

$$d_{jk} = \frac{\bar{x}_{T,jk} - \bar{x}_{N,jk}}{s_{p,jk}}.$$

The small-sample correction was applied:

$$J_{jk} = 1 - \frac{3}{4(n_{T,jk}+n_{N,jk}) - 9},$$

and cohort-specific Hedges' g was

$$g_{jk} = J_{jk} d_{jk}.$$

Its sampling variance and standard error were estimated as

$$\mathrm{Var}(g_{jk}) = \frac{n_{T,jk}+n_{N,jk}}{n_{T,jk}n_{N,jk}} + \frac{g_{jk}^2}{2(n_{T,jk}+n_{N,jk}-2)},$$

$$SE(g_{jk}) = \sqrt{\mathrm{Var}(g_{jk})}.$$

Positive g_{jk} values indicate higher expression in tumor than in normal mucosa, whereas negative values indicate lower expression in tumor. A cohort was included in the gene-level meta-analysis only when both g_{jk} and $SE(g_{jk})$ were finite and $SE(g_{jk}) > 0$.

Cross-cohort meta-analysis and directional consistency

For each gene j, valid cohort-specific Hedges' g estimates were combined using a random-effects meta-analysis fitted by restricted maximum likelihood when at least two valid cohort estimates were available. Let $\hat{g}_j$ denote the pooled Hedges' g, $SE(\hat{g}_j)$ its standard error, and p_j the corresponding two-sided meta-analytic P value. The 95% confidence interval, between-cohort variance τ_j^2, and I_j^2 statistic were retained for reporting.

A fixed-effect inverse-variance calculation was included only as a software fallback if random-effects fitting was unavailable. Random-effects meta-analysis was the intended primary estimator. The direction of the cohort-specific effects was also summarized. Let $K_{-,j}$ and $K_{+,j}$ denote the numbers of valid cohorts with $g_{jk} < 0$ and $g_{jk} > 0$, respectively, and let K_j denote the total number of valid cohort effects. The majority tumor-normal direction D_j and the proportion of cohorts supporting the majority direction C_j were defined as

$$D_j = \begin{cases} \text{Tumor\_down}, & K_{-,j} > K_{+,j}, \\ \text{Tumor\_up}, & K_{+,j} > K_{-,j}, \\ \text{Mixed\_or\_tie}, & K_{-,j} = K_{+,j}, \end{cases}$$

$$C_j = \frac{\max\left(K_{-,j}, K_{+,j}\right)}{K_j}.$$

Effects equal to zero contributed to K_j but to neither directional count.

Construction of the Meta-TN support score

The pooled result was converted to a bounded prioritization score using three quantities: the proportion of cohorts supporting the majority direction, the magnitude of the pooled standardized effect, and a *P*-value-derived measure of meta-analytic support.

The effect-magnitude component E_j was

$$E_j = \min\left(\frac{|\hat{g}_j|}{2}, 1\right),$$

and *P*-value-derived component P_j was defined as

$$P_j = \min\left(\frac{-\log_{10}\left[\max\left(p_j, \varepsilon\right)\right]}{10}, 1\right),$$

where ε denotes the smallest positive numerical value used to avoid evaluating $\log_{10}(0)$. Before applying the prespecified tumor-normal direction requirement, the base Meta-TN support score was defined as

$$S_j^{base} = C_j E_j P_j, \qquad 0 \le S_j^{base} \le 1.$$

The multiplicative form was used so that a high score required concurrent support from cross-cohort direction agreement, pooled effect magnitude, and the *P*-value-derived meta-analytic support component. The constants 2 and 10 were fixed scaling parameters used to place the magnitude and P-value-derived terms on bounded 0-1 ranges. Thus, $|\hat{g}_j| \ge 2$ gives the maximum magnitude contribution, and values with $p_j \le 10^{-10}$ receive the maximum *P*-value contribution. These constants were not estimated from GSE39582[1] or optimized against the external ranking, and their saturation points should not be interpreted as biological, clinical, or statistical decision thresholds. Because the meta-analytic *P* value depends on both effect magnitude and estimation precision, the magnitude and *P*-value-derived terms were not interpreted as independent evidence; the latter was used only as a bounded measure of meta-analytic support within the prioritization score.

Concordance with the prespecified tumor-normal direction

For the present 11-gene panel, the expected tumor-normal direction was defined from the same longitudinal multi-omics evidence used to establish the gene-specific StaBiCut direction priors. Accordingly, genes assigned as protective_high in this study were expected to be lower in tumor than in normal mucosa, whereas genes assigned as adverse_high were expected to be higher in tumor. This mapping was specific to the direction-prior scheme used for the present candidate panel and should not be interpreted as a general equivalence between prognostic direction and tumor-normal expression change. Let I_j^{prior}= 1 when D_j matched this expected tumor-normal direction and I_j^{prior} = 0 otherwise:

$$I_j^{prior} = \begin{cases} 1, & D_j = D_j^{expected}, \\ 0, & otherwise, \end{cases}.$$

The prior-concordant support score was

$$S_j^{prior} = I_j^{prior} S_j^{base}.$$

For integration with GSE39582[1], a final concordance check compared the sign of the pooled Hedges' g with the high-versus-low hazard direction observed at the selected external cutoff. Let HR_j^{ext} denote the hazard ratio for the high-expression group relative to the low-expression group, and define

$$I_j^{HR} = \boldsymbol{I}\{\mathrm{sign}(\hat{g}_j)\mathrm{sign}[\log(HR_j^{ext})] = 1\}.$$

The Meta-TN component entering the extended composite score was

$$S_j^{MetaTN} = I_j^{HR} S_j^{prior}.$$

Accordingly, a positive pooled Hedges' g was required to accompany an adverse high-expression association in GSE39582, whereas a negative pooled Hedges' g was required to accompany a protective high-expression association. Otherwise, the Meta-TN component was set to zero. Because the GSE39582 cutoff scan had already been constrained by the same gene-specific hazard-direction priors, this step served only as a final consistency check and did not alter the selected cutoff or the external survival model.

Tumor-only and Meta-TN-extended external composite scores

Let $S_{\mathrm{boot},j}$, $S_{\mathrm{density},j}$, $S_{\mathrm{hazard},j}$, and $S_{\mathrm{balance},j}$ denote the GSE39582 bootstrap-stability, cutoff-plausibility, bootstrap hazard-direction consistency, and group-balance scores, respectively. When the tumor–normal component was unavailable, the tumor-only external composite score was normalized by the sum of the retained original weights:

$$S_j^{no\ TN} = \frac{0.40S_{\mathrm{boot},j} + 0.20S_{\mathrm{density},j} + 0.15S_{\mathrm{hazard},j} + 0.10S_{\mathrm{balance},j}}{0.85}.$$

The tumor-only score and its corresponding rank were calculated in the upstream GSE39582 StaBiCut runner. The Meta-TN extension read these previously generated values and did not reconstruct or alter the four

GSE39582-derived component scores. After incorporation of the external Meta-TN component, the original five-component weight structure was reinstated:

$$S_j^{ext+MetaTN} = 0.40S_{\text{boot},j} + 0.20S_{\text{density},j} + 0.15S_{\text{hazard},j} + 0.10S_{\text{balance},j} + 0.15S_j^{MetaTN}.$$

Genes were ranked in descending order of the corresponding composite score; genes with identical scores received the same rank. Addition of the external Meta-TN component affected only the final Meta-TN-extended composite score and ranking. The selected GSE39582 cutoff and all four GSE39582-derived component scores remained unchanged.

References

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